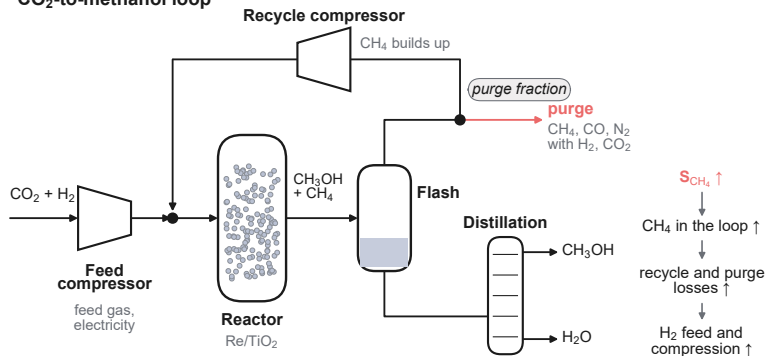
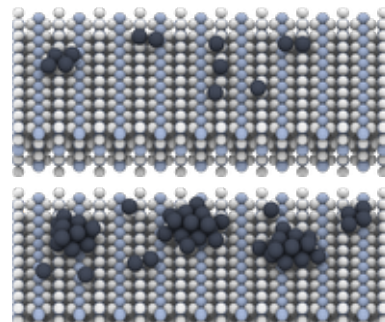


a**CO₂-to-methanol loop****b****Re/TiO₂ catalyst states (schematic)**

measured at 200 and 250 °C, 100 bar,
CO₂ : H₂ = 1 : 4

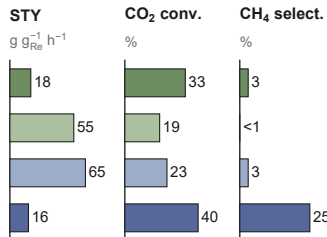
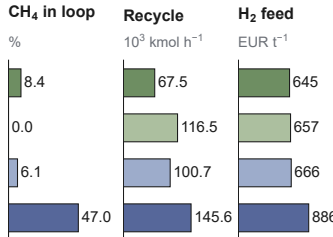
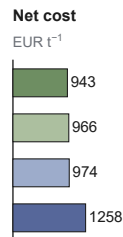
1 wt% Re
12 Re atoms

5 wt% Re
60 Re atoms

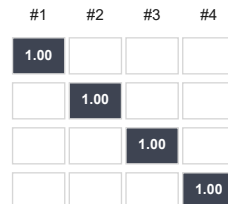
**c****Catalyst, measured**

Catalyst state
in net-cost order

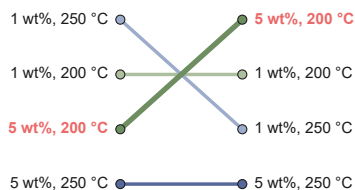
■ 5 wt%, 200 °C
■ 1 wt%, 200 °C
■ 1 wt%, 250 °C
■ 5 wt%, 250 °C

**Loop at 2% purge****Plant****d Rank probability**

5,000 cost draws

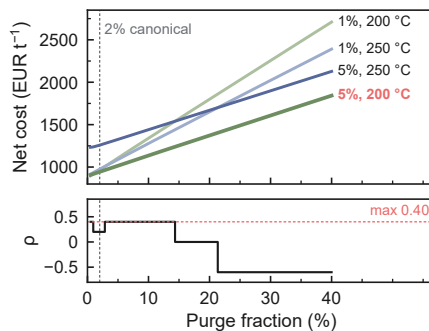


canonical and active-Re
boundaries, 5,000 / 5,000

e**STY per g Re Net cost**

$\rho = 0.20$, $\tau = 0.00$

3 of 6 pairs inverted

f**g**